

EXAM IN: STA510 STATISTICAL MODELING AND SIMULATION

DURATION: 4 HOURS

DATE: NOVEMBER 17th, 2025

PERMITTED AIDS: Approved simple calculator

One A4 size sheet with your own handwritten notes.

THE EXAM CONSISTS OF 2 PROBLEMS ON 6 PAGES, 6 PAGES OF ENCLOSURES.

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Problem 1

We like to model and analyze the rainfall in Stavanger during a storm. The English term storm describes here a low-pressure area with some rain, and a storm event is defined hydrologically as a continuous precipitation episode separated from others by ≥ 6 hours of no rain. Let X be a random variable describing the *rainfall depth per storm* (*mm per m²*). We assume that the rainfall depths for individual storms are independent and identically distributed as

$$X \sim \text{Exp}(\beta) \quad \text{with} \quad f_X(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x \geq 0.$$

Assume further, we have collected $n = 40$ storm depths over a recent period:

2.5	4.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0	13.0
5.0	13.0	7.0	14.0	9.0	11.0	8.0	10.0	6.0	12.0
3.0	5.0	9.0	10.0	8.0	7.0	15.0	14.0	9.0	12.0
6.0	13.0	11.0	8.0	10.0	9.0	16.0	5.0	7.0	17.0

with sample mean of $\bar{X} = 9.29\text{mm}$ and sample standard deviation of $\sigma = 3.53\text{mm}$.

- a) Assuming a scale parameter of $\beta = 8\text{mm}$ for Stavanger.
 - i) Find the probability that it rains more than 12mm during a single storm event in Stavanger. (3P)
 - ii) Given that it has already rained 5mm in a storm, find the probability that the storm's total rainfall exceeds 12mm . (3P)
- b) Now, we like to estimate β from the data.
 - i) Show the the Likelihood function is given by $L(\beta) = \beta^{-n} e^{-\frac{1}{\beta} \sum_{i=1}^n x_i}$. (2P)
 - ii) Show that the Maximum Likelihood estimator (MLE) is $\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i$. (4P)
 - iii) Is $\hat{\beta}$ an unbiased estimator? Reason your answer! (2P)
 - iv) Show that the variance of $\hat{\beta}$ is $\frac{\beta^2}{n}$. (2P)
- c) How many measurements of the rain depth should we take to be 95% sure that $\hat{\beta}$ falls at most a margin of 1mm from β ? Reason your answer! (3P)

- d) Now, we want to draw $B = 4000$ non-parametric bootstrap samples from the 40 rainfalls depth above. Given the R-code and output below:
- Estimate the bias and the MSE for $\hat{\beta}$, and calculate the bias corrected estimate. (6P)
 - Construct a 95% standard normal bootstrap interval, 95% basic bootstrap interval, and 95% percentile bootstrap interval. (8P)

```
rain <- c(2.5,4.0,6.0,7.0,8.0,9.0,10.0,11.0,12.0,13.0,
          5.0,13.0,7.0,14.0,9.0,11.0,8.0,10.0,6.0,12.0,
          3.0, 5.0, 9.0, 10.0, 8.0, 7.0, 15.0, 14.0, 9.0, 12.0,
          6.0, 13.0, 11.0, 8.0, 10.0, 9.0, 16.0, 5.0, 7.0, 17.0 )
n <- length(rain)

B <- 4000
estB <- numeric(B)
set.seed(123)
for(i in 1:B)
  estB[i] <- mean(sample(rain,size=n,replace = TRUE))

mean(estB)

## [1] 9.277269

sd(estB)

## [1] 0.5502607

# Some quantiles for normal distribution
my.quant1 <- c(0.01,seq(0.025,0.1,0.025))
print(t(data.frame(Probability = my.quant1,
                   Quantile = round(qnorm(my.quant1),2))))

##           [,1]    [,2]    [,3]    [,4]    [,5]
## Probability 0.01  0.025  0.05  0.075  0.10
## Quantile   -2.33 -1.960 -1.64 -1.440 -1.28

# Some quantiles for bootstrap sample
my.quant2 <- c(seq(0,0.05,0.025), seq(0.1,0.9,0.2), seq(0.95,1,0.025))
round(quantile(estB, my.quant2),2)

##      0%  2.5%   5%   10%   30%   50%   70%   90%   95% 97.5% 100%
##  7.39  8.18  8.38  8.57  8.99  9.28  9.57  9.99 10.20 10.36 11.10
```

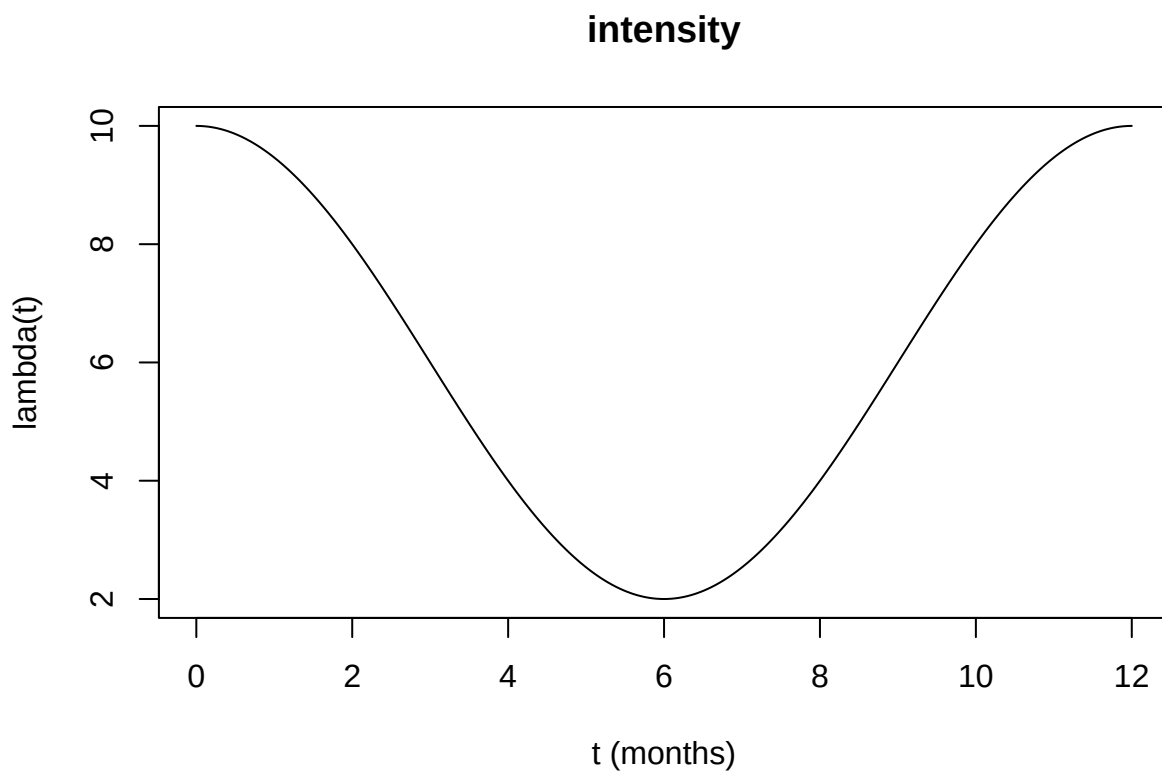
Assume storms follow a homogeneous Poisson process with rate $\lambda = 6$ storms/month. Let $N(t)$ be the number of storms in t months.

- e) i) What are the expected number of storms and the variance of the number of storms during any t -month period? (2P)
- ii) We are interested to estimate the probability that Stavanger experiences 90 or more storms in one year. Describe how we could simulate the numbers of storm in one year, and how we could estimate the probability to have 90 or more storms in one year. (6P)

- f) Let us assume we want to model the number of storms as a non-homogenous poisson process. For this we consider a intensity $\lambda(t)$ as shown below.
- i) What is the interpretation of the intensity function? (3P)
 - ii) Does homogenous or non-homogenous poisson process fits better to model the number of storms? Reason your answer! (3P)
 - iii) Describe how to model a non-homogenous poisson process given the intensity $\lambda(t)$ as shown below. (5P)

```
lambda <- function(t) 2 + 8 * (sin(pi * (t-6) / 12))^2  
tt <- seq(0,12,length.out=1000)
```

```
plot(tt, lambda(tt), type = "l", xlab = "t (months)", ylab = "lambda(t)",  
     main = "intensity")
```



Suppose historical weather records for Stavanger suggest that the average rainfall per storm is typically around $8mm$, but with some uncertainty. We want to update this belief using the new observed rainfall data (the 40 depths from the dataset above). Assume a prior distribution $\beta \sim \text{Gamma}(a_0 = 4, b_0 = 2)$ with prior mean of $8mm$.

g) Show that the log-posterior density kernel $\log(g(\beta))$ is given by

$$\log(\pi(\beta|x)) \propto \log(g(\beta)) = (a_0 - n - 1) \log(\beta) - \frac{\beta}{b_0} - \frac{\sum x_i}{\beta}. \quad (4P)$$

h) Below we see R-code with an implementation of a MCMC algorithm. What kind of MCMC algorithm is this, and how is the performance of the algorithm given the R-output? Why is the below implementation not suitable for the target distribution? Reason your answers! (6P)

```
a0 <- 4
b0 <- 2
sumX <- sum(rain)
n <- length(rain)

sigma <- 4.0
n.iter <- 20000
theta1 <- 6

# posterior target
lpost <- function(beta){
  if(beta>0)
    gbeta <- (a0-n-1)*log(beta) - beta/b0 - sumX/beta
  else
    gbeta <- - 1e100
  return(gbeta)
}
set.seed(123)
output <- numeric(n.iter)
output[1] <- theta1
acc_mh <- 0
for(t in 2:n.iter){
  thetaStar <- rnorm(1,mean=output[t-1],sd=sigma)
  alpha <- exp(min(0.0,lpost(thetaStar)-lpost(output[t-1])))
  # accept/reject step
  if(runif(1)<alpha && is.finite(alpha)){
    output[t] <- thetaStar
    acc_mh <- acc_mh + 1
  } else {
    output[t] <- output[t-1]
  }
}
```

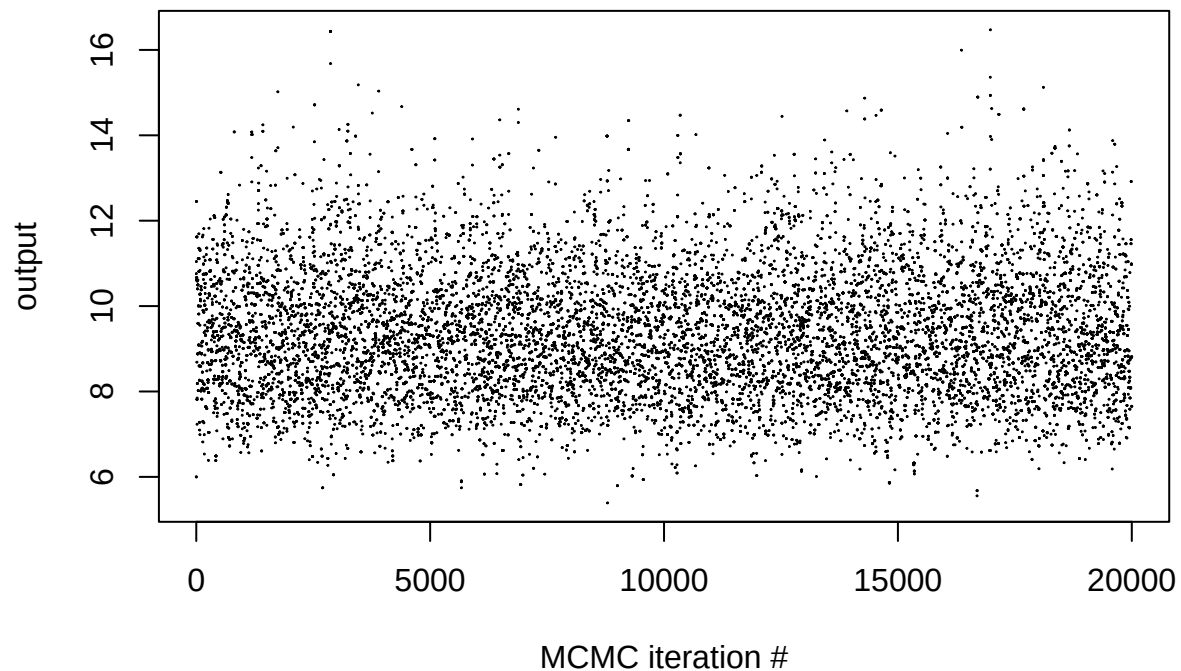
```
# Acceptance rate
acc_mh / (n.iter-1)
```

```
## [1] 0.3723186
```

```
# Effective sample size
coda::effectiveSize(output)
```

```
##      var1
## 4088.635
```

```
plot(1:length(output),output,pch=20,cex=0.1,xlab="MCMC iteration #")
```



Now we want to improve the MCMC algorithm for estimating the posterior.

- i) Suggest a log-scale Metropolis-Hasting proposal with $\log(\beta^*) = \log(\beta_{t-1}) + \varepsilon$ and $\varepsilon \sim N(0, \sigma^2)$. Find the proposal density $q(\cdot | \beta_{t-1})$ for the proposal mechanism, and describe how to compute the acceptance probability $\log(\alpha)$. (6P)
- j) Suggest an independent Gamma proposal distribution that is centred near the MLE $\hat{\beta}$, i.e. with mean $\hat{\beta}$ and variance near the estimated sample variance of $\hat{\beta}$. Find the proposal density $q(\cdot)$ including parameters, and describe how to compute the acceptance probability $\log(\alpha)$. (6P)

Problem 2

Assume we want to estimate the integral

$$I = \int_0^\infty \ln(1 + 4x^2)e^{-\frac{1}{2}x} dx.$$

- a) Write down an expression for a Monte Carlo estimator of I based on n random draws from an exponential distribution. In addition, describe the steps involved in constructing and using this estimator. (4P)
- b) Show that the inverse cumulative distribution function of the Exponential distribution with scale parameter β is given as $F_X^{-1}(U) = -\beta \ln(U)$. (4P)
- c) Write down an expression for an antithetic Monte Carlo estimator of I based on n random draws. In addition, describe the steps involved in constructing and using this estimator. Do you think the antithetic Monte Carlo estimator works well for I ? Why? (6P)